

LOGGY CONTROL BOARD: ENGG3800 TEAM6 2025 Sem2

MCU PINOUT AND PROGRAMMING

SS1
MCU_PROGRAMMING.SchDoc



MCU Control and programming header. Details communication protocols and pinouts for MCU(U2:STM32L433CCT6 - 48LQFP).

ANALOG CHANNELS (CH1-CH4)

SS2
CH1_4_INPUT_CONDITIONING.SchDoc



Full analog signal flow from input terminal blocks (J1-J4) to signal input to ADC (U4: ADS1119).

DIGITAL CHANNELS (CH5-CH8)

SS3
SENSORS_5-8.SchDoc



Accelerometer (U3:LISD3H) and Temperature sensing (ONBOARD_TEMP).

VOLTAGE SOURCES (±12V, ±5V, +3.3V, +9V)

SS4
VOLTAGE_SOURCES_RAILS.SchDoc



All sources powered from 6xAA 1.5V batteries for nominal +9V supply. ±12V, +5V, +3.3V powered directly from battery. -5V from +5V source.

ANALOG TO DIGITAL CONVERTER (ADC)

SS5
ADC.SchDoc



U18:ADS1119 converts single analog input to digital output. Communicates via I2C with MCU (U2). Reference voltage to +3.3V rail. 16-bit resolution over FSR from 0V to +3.3V.

TOSLINK TRANSCEIVER

SS6
TOSLINK.SchDoc



Utilises U12: MAX941CPA+ to interface with control board via TOSLINK cable. UART protocol with through-hole transmit LED and BPW34 receive photodiode.

CURRENT SOURCES (10uA AND 200uA)

SS7
CURRENT_SOURCES.SchDoc



Utilises two LM334Z chips (U4&U5) to generate 10uA and 200uA current sources. Sources are available at screw terminal block (J3).

ALARM INDICATORS

SS8
LED_CONTROLLER.SchDoc



16-bit constant-current LED sink driver (U17:TLC5927IDWR) controls ~10mA across 16 surface mount RED LEDs.

TEMPERATURE SENSORS

SS9
TEMPERATURE_SENSORS.SchDoc



Contains temperature sensors (thermistor and RTD) and respective wire details.

MECHANICALS

SS7
Mechanicals.SchDoc



Mechanical and mounting hardware. Fiducials and testpoint.

Title		
Sheet Symbol Title Page (SS1:SS9)		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 0 of 10
File:	X:\Files\LANDING PAGE.SchDoc	Drawn By: TEAM6

STM32L433CCT6 - 48LQFP COMMUNICATION AND PINOUT

Accelerometer (U3)
communication via
I2C1 with MCU (U2)

+3V3 100nF, 50V, X7R, ±10%

C1

GND

C1,C2,C5,C6 decoupling
capacitors must be placed
as close to U1 (VDD PINS)
as possible

R32 sets LED24
current at 12.2mA
LED24 indicates
sampling occurring
(active when sampling
active).

LED24
2.5V, 25mA, RED

R32

270Ω, ±5%, 1/8W

GND

HDR_RST

ONBOARD_NTC

TOSLINK_TX

+3V3

C5

100nF, 50V, X7R, ±10%

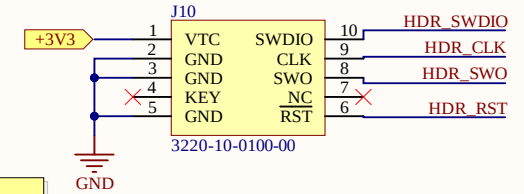
ONBOARD_NTC signal
(PA0:Pin10) is backup
onboard temperature
reading if on-chip
measurement is inaccurate.

Accelerometer (U3) uses I2C (I2C1).
ADC uses I2C (I2C3), LED control
(U17) uses SPI2 and GPIO PB14,
PB12. MCU (U2) reset controlled by
SWD.

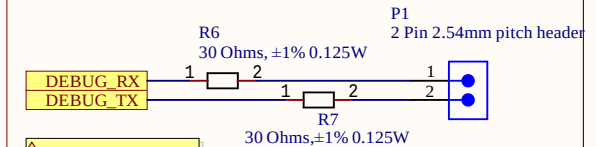
ADC (U18:ADS1119)
communication via I2C3
with MCU (U2).

PB0:PB2 allow control of MUX
(U23) selection. Configured as
GPIO pins, allows 3 bit selection
from 8 channels(CH1:4, 10V & 1V
ranges) to 1 (signal 'MUX_OUT')

SWD 10-PIN PROGRAMMING HEADER



UART DEBUG PINS

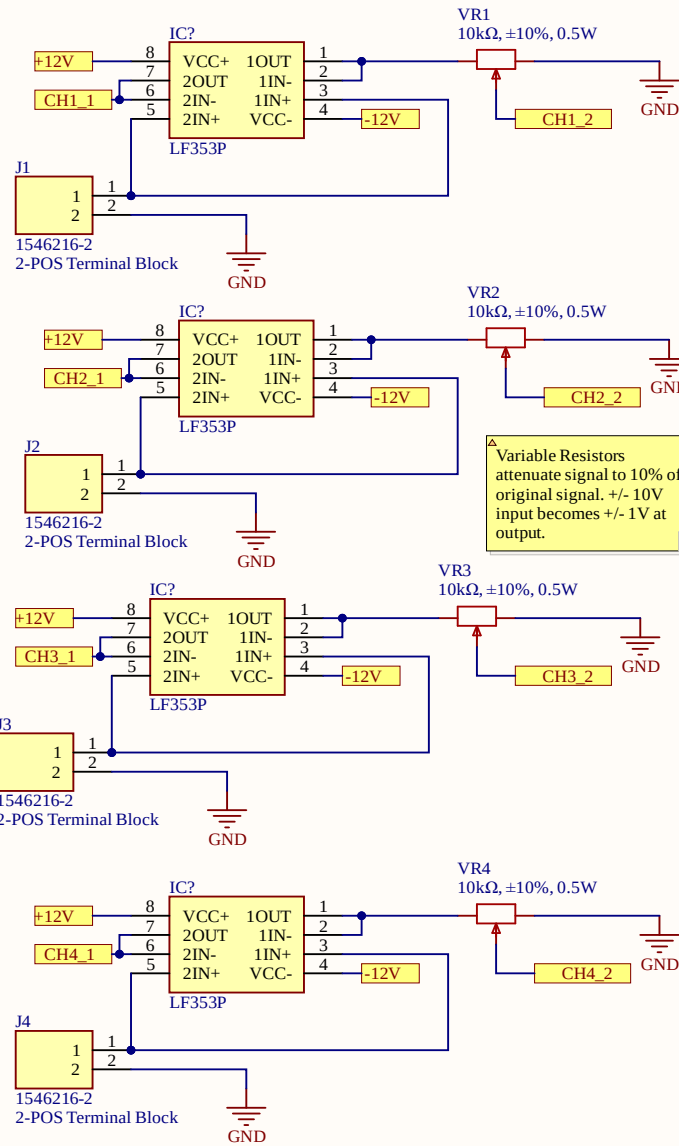


DEBUG_RX/TX
(PA10/PA9) exposed
via two pin header for
debugging purposes.
Provides monitoring
over USART1.

LED controller
communication is via
SPI2 (PB15, PB13) for
MOSI and CLK
respectively. PB12 and
PB14 are configured as
GPIO for LE and OE
control respectively.

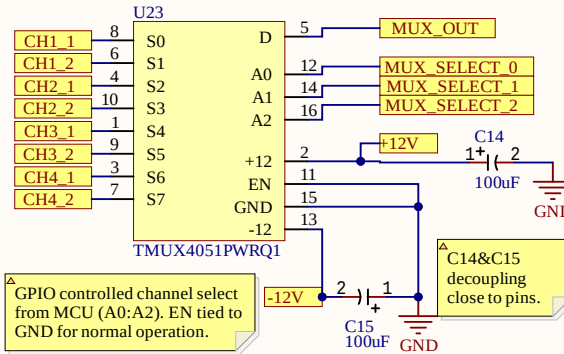
Title		
MCU AND PROGRAMMING		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 1 of 10
File:	C:\Users\...MCU_PROGRAMMING.Sch	Drawn By: TEAM6

CH1-4 INPUT SCREW TERMINAL BLOCKS AND BUFFERS

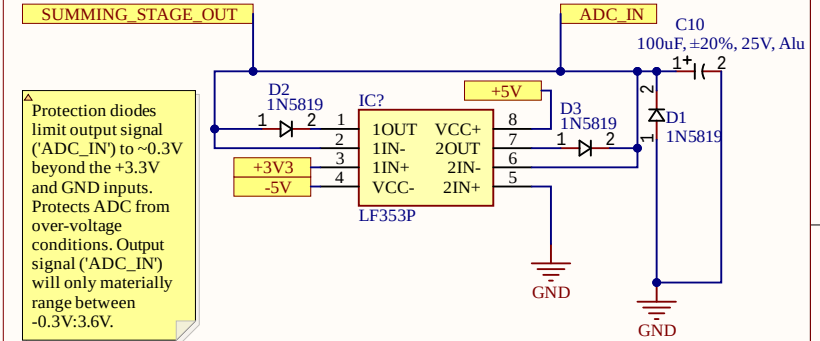


CH1-4 INPUT 8:1 MUX

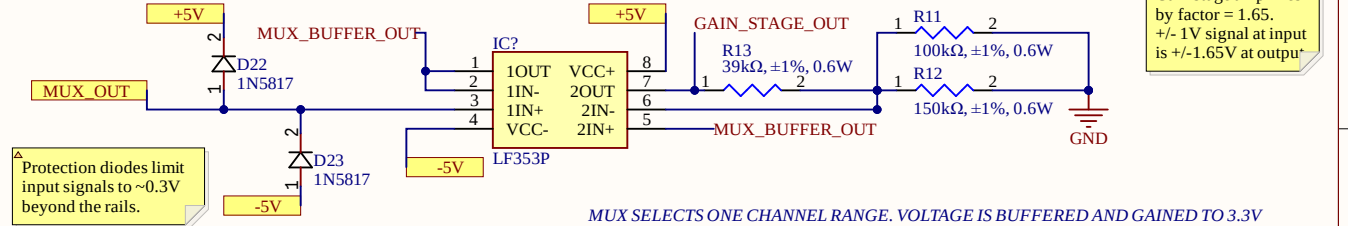
2 RANGES (+/- 1V AND +/- 10V) PER CHANNEL, MCU SELECTS ONE



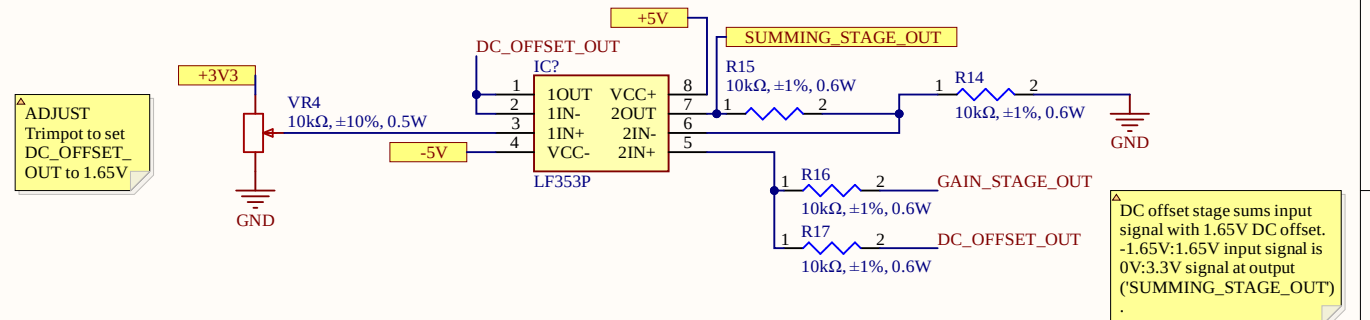
ADC INPUT BUFFER AND PROTECTION



MUX OUTPUT VOLTAGE BUFFER AND GAIN OUTPUT

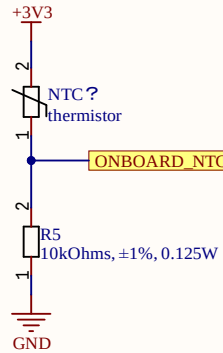


DC OFFSET ON ADC INPUT SIGNAL



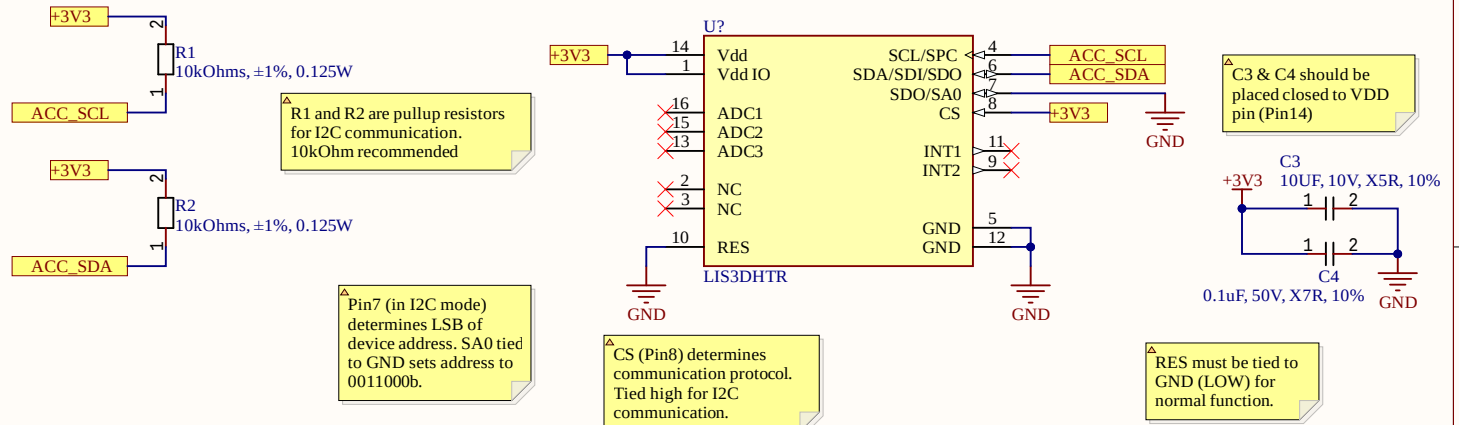
Title		
SIGNAL CONDITIONING FROM INPUT TO ADC (CH1:CH4)		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 2 of 10
File:	X:\Files\CH1_4_INPUT_CONDITIONING.DWG	TEAM 6

ONBOARD TEMPERATURE SENSOR (CH8)



Provides backup temperature measurement in even that onchip (U3) temperature measurement exhibits too much self heating to be accurate.

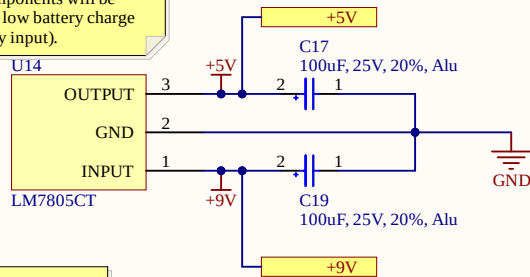
ONBOARD 3-AXIS ACCELEROMETER (X=CH5, Y=CH6, Z=CH7)



Title		
ACCELEROMETER AND TEMPERATURE CHANNELS (CH5:CH8)		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 3 of 10
File:	X:\Files\...\SENSORS_5-8.SchDoc	Drawn By: TEAM 6

5V SOURCE FROM BATTERY

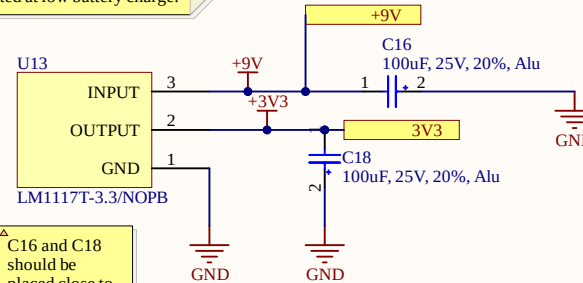
U14 has 2V dropout, 1.5A supply at fixed 5V. Supply to digital components will be affected at low battery charge (7V battery input).



C17 and C19 should be placed close to chip U14

3.3V SOURCE FROM BATTERY

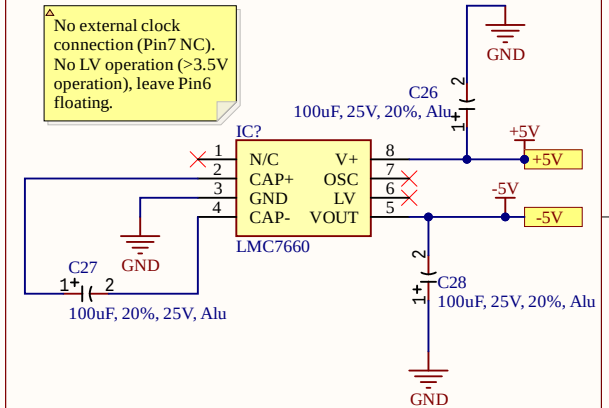
U13 has 1.2V dropout, 800mA supply at 3.3V fixed. Supply to digital components will be affected at low battery charge.



C16 and C18 should be placed close to chip U13

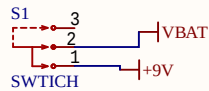
-5V SOURCE FROM +5V SUPPLY

No external clock connection (Pin7 NC). No LV operation (>3.5V operation), leave Pin6 floating.



Battery Power Switch

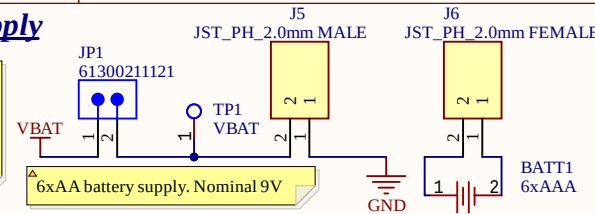
Controls nominal 9V supply to board



When switch is towards input terminals (J1-5), battery is connected to board.

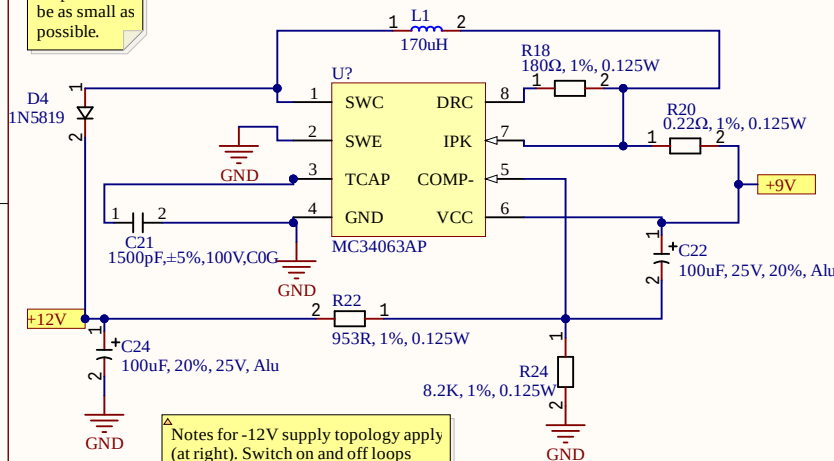
Battery Supply

JP1 is a two pin header for current monitoring from battery. Must be shunted for board to have power.



+12V SUPPLY RAIL

Inductor and switching loops should be as small as possible.

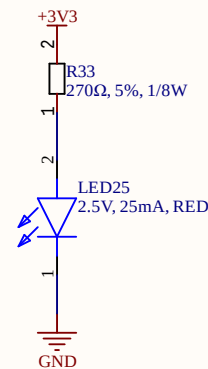


Notes for -12V supply topology apply (at right). Switch on and off loops should be small, C21 should be close to pins, input and output capacitors should be close to pins/small loops

+3V3 RAIL INDICATOR LED

LED25 (RED) will be on when rail is active.

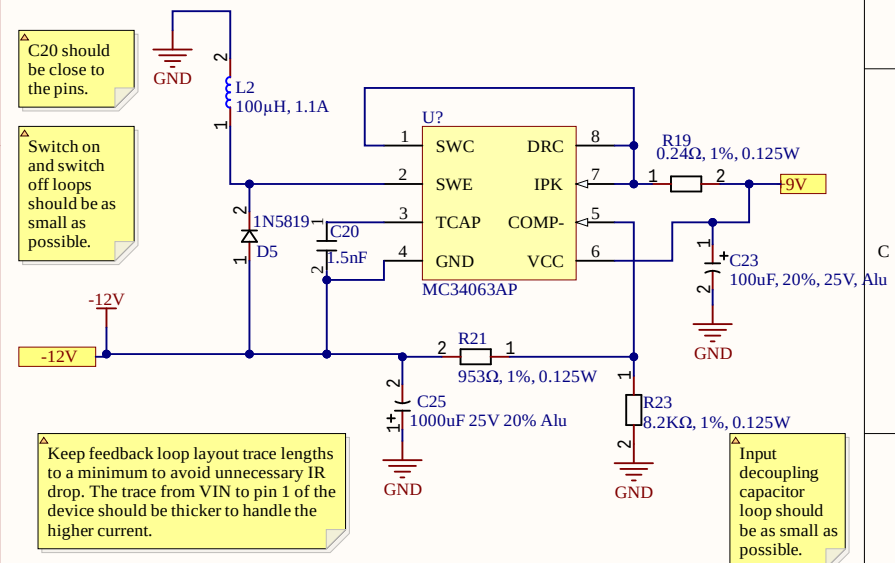
R33 sets LED25 current at 12.2mA



-12V SUPPLY RAIL

C20 should be close to the pins.

Switch on and switch off loops should be as small as possible.

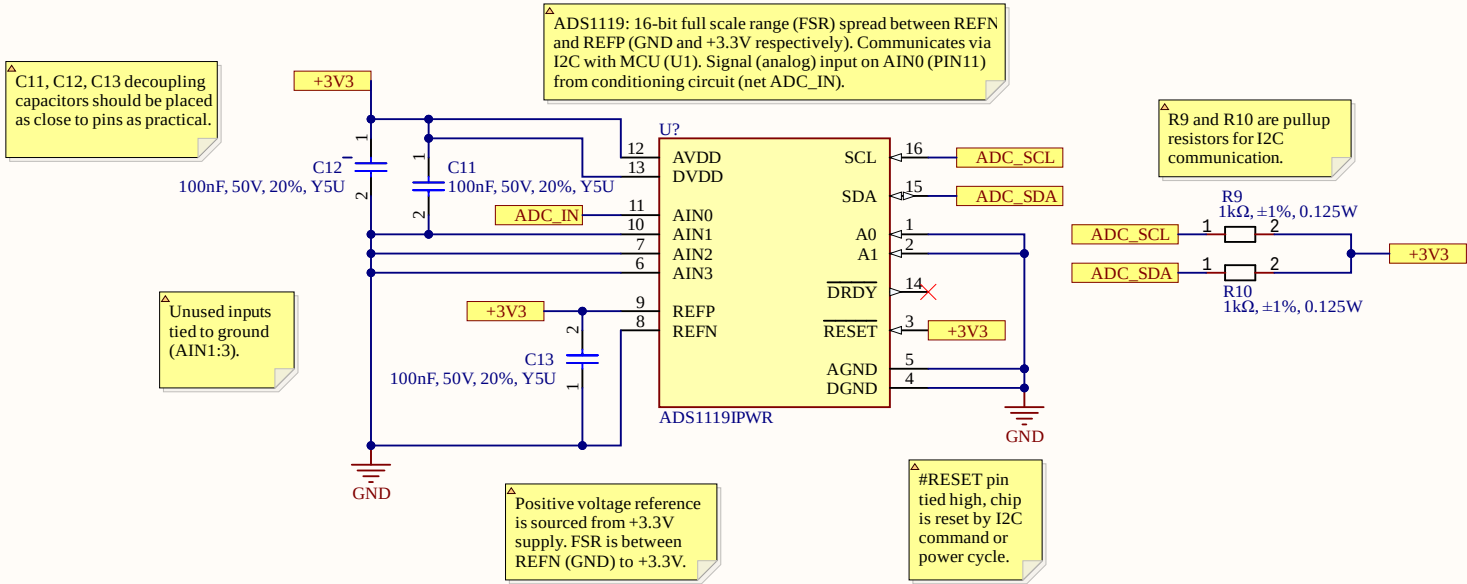


Keep feedback loop layout trace lengths to a minimum to avoid unnecessary IR drop. The trace from VIN to pin 1 of the device should be thicker to handle the higher current.

Input decoupling capacitor loop should be as small as possible.

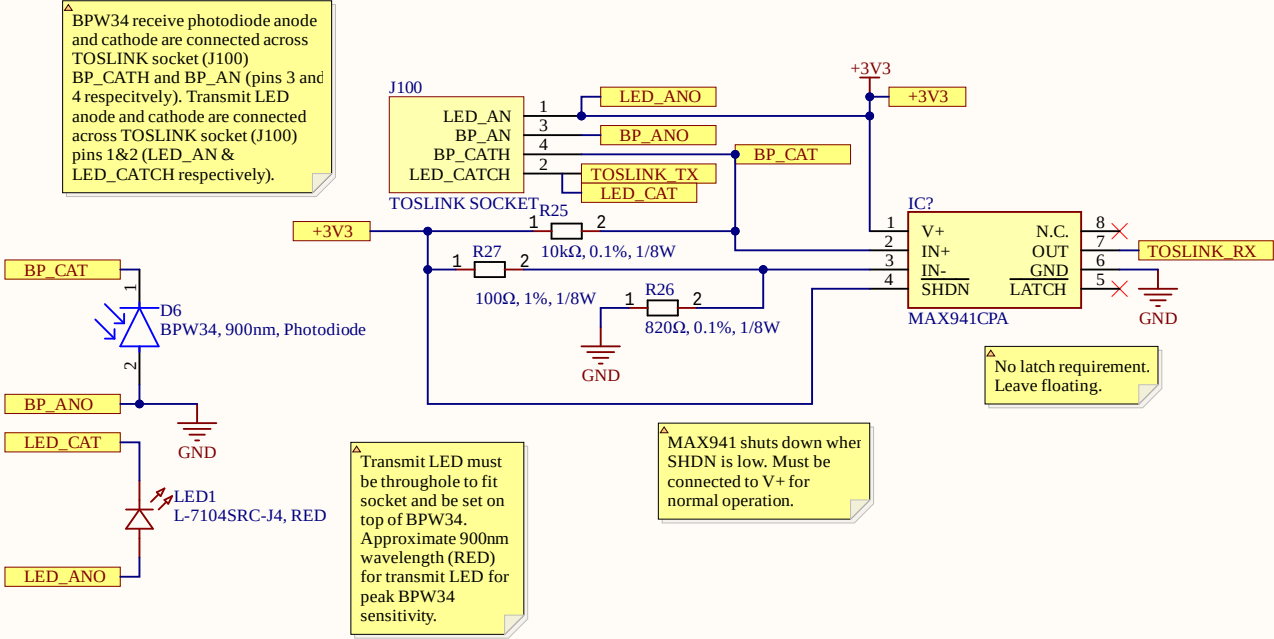
Title		
VOLTAGE SOURCES AND RAILS		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 4 of 10
File:	X:\Files\...VOLTAGE_SOURCES_RAILS_Sch.Dwg	TEAM 6

ADC (ADS1119)



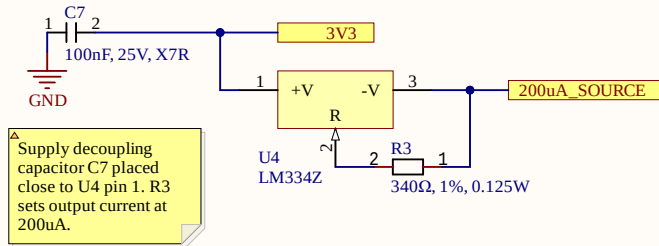
Title		
ANALOG TO DIGITAL CONVERTER (ADS1119)		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet5 of 10
File:	X:\Files\...\ADC.SchDoc	Drawn By: TEAM6

TOSLINK RECIEVE AND TRANSMIT

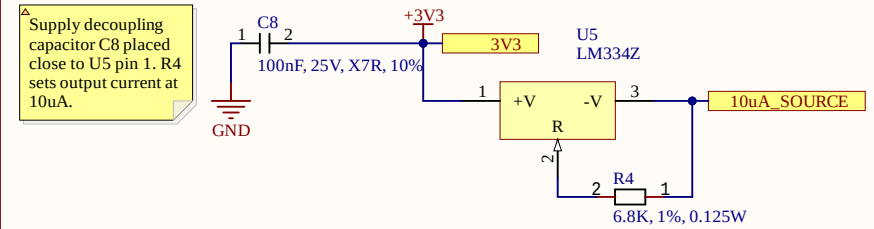


Title		
TOSLINK Transceiver		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 6 of 10
File:	X:\Files\TOSLINK.SchDoc	Drawn By: TEAM 6

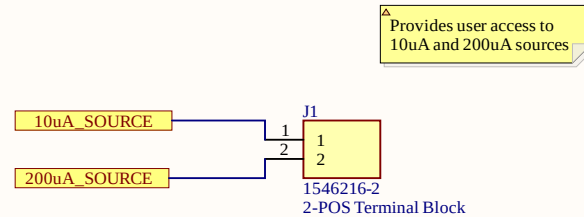
200uA Current Source



10uA Current Source

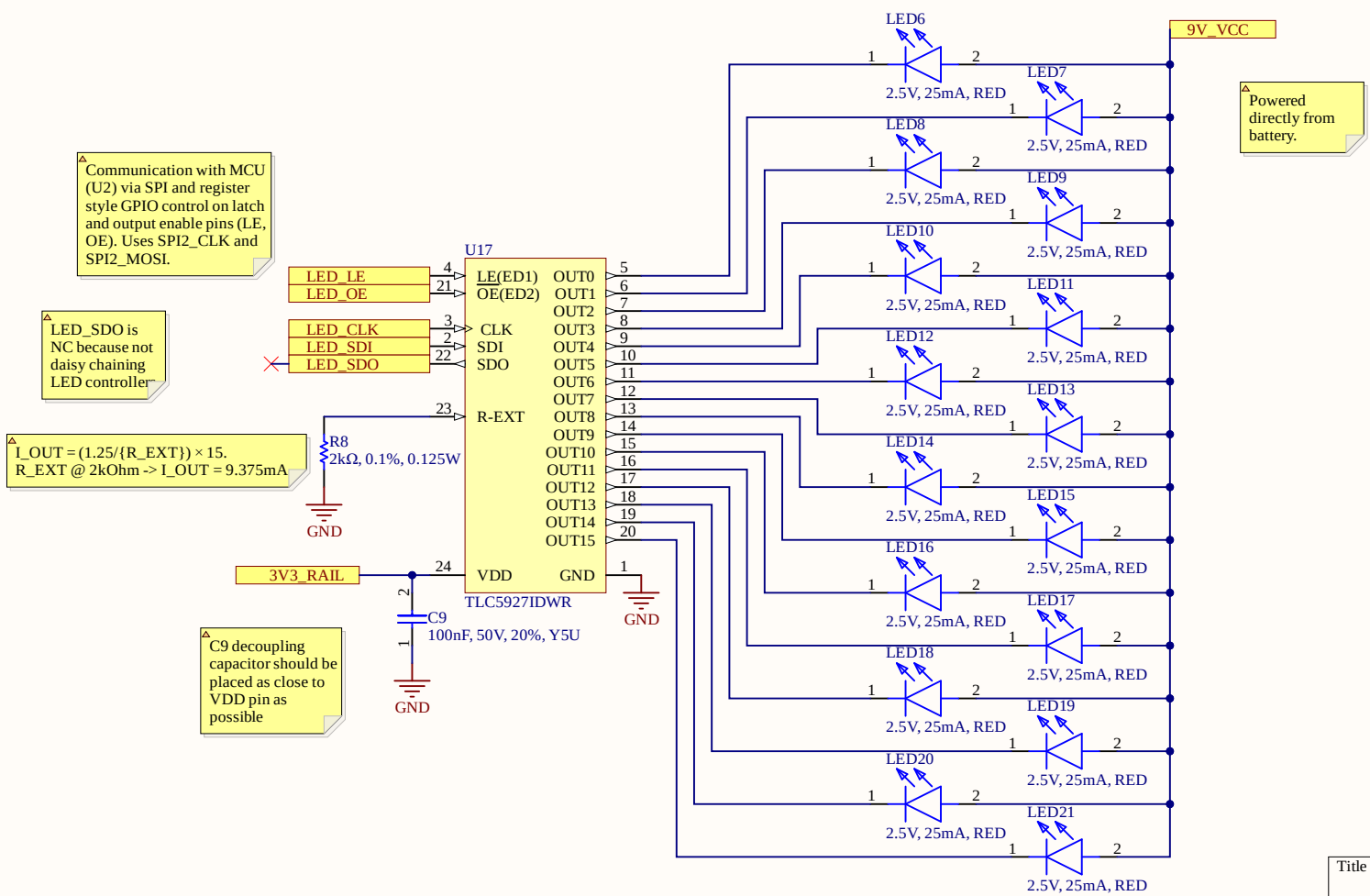


Current Sources Terminal Block Configuration



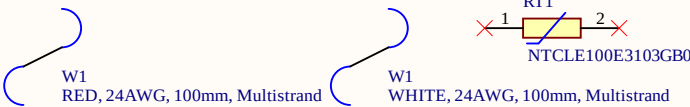
Title		
10uA and 200uA CURRENT SOURCES		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 7 of 10
File:	X:\Files\...\CURRENT SOURCES.SchDoc	Drawn By: TEAM 6

ALARM LEDS: 16-bit constant-current LED sink driver



Title		
ALARM LED CONTROL		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 8 of 10
File:	X:\Files\LED CONTROLLER.SchDoc	Drawn By: TEAM 6

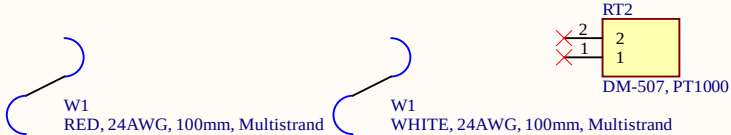
TEMPERATURE SENSOR: THERMISTOR



NTCLE100E3103GB0 : NTC
10KOHM 3977K BEAD

Supplied with 100mm, 24AWG, multistrand wire connected to legs of thermistor. Yellow heatshrink, RED wire for leg/pin 1, GREEN heatshrink, WHITE wire for leg/pin 2. Terminal ends are tinned for insertion into screw terminal.

TEMPERATURE SENSOR: PLATINUM RTD (PT1000)



Supplied with 100mm, 24AWG, multistrand wire connected to legs of RTD. Yellow heatshrink, RED wire for leg/pin 1, GREEN heatshrink, WHITE wire for leg/pin 2. Terminal ends are tinned for insertion into screw terminal.

Title			TEMPERATURE SENSORS
Size	Number	Revision	
A4	1	1	
Date:	5/26/2025	Sheet 9 of 10	
File:	X:\Files\TEMPERATURE SENSORS	Submitted By:	TEAM 6

1

2

3

4

MOUNTING HOLES

STANDOFFS

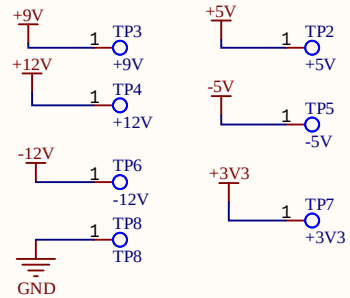
Standoffs
comprised of
pillar and nut

VELCRO

Approx 100mm of
Velcro tape is
required to secure
battery case to
underside of
sensor board PCB

FIDUCIALS

VOLTAGE SUPPLY TESTPOINTS



Title		
MECHANICALS		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 1 of 10
File:	X:\Files\...\Mechanicals.SchDoc	Drawn By: TEAM6

1

2

3

4